

AI Engineer Syllabus

- Duration: 3 Hours
- Format: Lecture + Live Coding + Hands-on Build
- Audience: Engineering Students
- Prerequisites: Basic Python, familiarity with REST APIs

Tech Stack

Python, FastAPI, Gemini API, ChromaDB, an open-source embedding model

Pre-Session Setup

- Python 3.10+, VS Code
- Google account with a Gemini API key generated in advance
- GitHub account

Session Breakdown

Segment	Duration
Introduction to Modern AI	15 min
Foundations of Generative AI	15 min
Prompt Engineering	20 min
Project 1: ChatBot Agent	55 min
Break	10 min
Project 2: RAG	50 min
Wrap-up & Q&A	15 min
Total	180 min

Module 1: Introduction to Modern AI

- Evolution of software: rule-based systems, machine learning, large language models, AI agents
- AI application architecture: client layer, API layer, orchestration layer, model layer, data layer
- The AI development lifecycle: prototype, evaluate, harden

Module 2: Foundations of Generative AI

- Large language models and autoregressive text generation
- Tokens and context windows
- Multimodal AI: text, image, video
- Embeddings and semantic similarity
- Structured output
- Tool calling
- Retrieval-Augmented Generation
- Model Context Protocol

Module 3: Prompt Engineering

- Prompt design principles: clarity, specificity, role definition, instruction ordering
- System prompts versus user prompts
- Few-shot prompting
- Chain-of-thought prompting
- XML-tag structured prompts
- Prompt templates

Project 1: ChatBot Agent

Phase 1 — Prompt Design + Basic Chat

- System prompt design: persona, scope, tone
- FastAPI backend: request/response structure, validation
- A working /chat endpoint calling the Gemini API

Phase 2 — Tool Calling

- The agent loop: plan, act, observe
- Gemini function calling
- Tool schema design
- Three tools: date/time lookup, calculator, FAQ lookup
- The model autonomously selecting which tool to call

Project 2: RAG (Retrieval-Augmented Generation)

Phase 1 — Ingest & Embed

- Document loading and chunking
- Generating embeddings with an open-source embedding model
- Storing embeddings in a persistent ChromaDB collection

Phase 2 — Retrieve & Generate

- Embedding an incoming query
- Similarity search against the vector store
- Constructing a grounded prompt from retrieved context
- Generating an answer with source citations
- Correctly declining to answer when the knowledge base has no relevant information

Learning Outcomes

- Design effective prompts using established prompt engineering techniques
- Build a conversational AI backend with FastAPI
- Implement tool calling to give an AI application the ability to take action
- Build a complete RAG pipeline from document ingestion to grounded generation

Additional Topics for Discussion

- Small Language Models (SLMs)
- Project Deployment: Docker, Kubernetes, Agent Platforms, GCP
- Model Deployment: Self-Hosted Models
- Cost Optimization: Token Usage, Model Selection, Caching
- Security & Guardrails: Prompt Injection, Data Privacy, Output Filtering
- Multi-Agent Systems
- AI Evaluation & Benchmarking
- Harness Engineering
- Fine-Tuning versus RAG